which are active-low. As the counter changes, the decoder selects AP1, AP2, AP3, and so on up to AP16.

For example, the count 0011 selects AP4, and pin 4 of IC3 goes low. LED2 to LED5 show the binary count so you can see which channel is selected (see the table).

Relay and appliance switching unit. This part uses 16 BC557 transistors (T1-T16) as driver stages, 16 SPDT relays (RL1-RL16) and 16 flyback diodes (D1-D16). Each decoder output drives its own transistor, which energises its 5V relay. A 1N4007 diode across each relay coil protects the transistor from the voltage spike produced when the relay switches off. The relay contacts isolate the 5V control circuit from the 230V AC load, and each relay can switch its appliance within the relay's rated capacity.

Using S1 and S2, you select one appliance at a time, and LED2 to LED5 indicate the selection. The table lists the corresponding output status.

BILL OF MATERIALS

Reference	Component	Value/Type	Quantity	Cost (₹)
IC1 (IC1A-IC1D)	Quad 2-input NAND Schmitt trigger	CD4093	1	33
IC2	Presettable up/down counter	CD4029	1	43
IC3	4-line-to-16-line decoder/demultiplexer	74LS154	1	120
T1-T16	PNP transistor	BC557	16	29
RL1-RL16	Relay	5V SPDT	16	640
LED1-LED5	LED	5mm	5	Miscellaneous (~235)
D1-D16	Rectifier diode	1N4007	16	
R1-R3, R9-R24	Resistor	1kΩ, ¼W, 5%	19	
R4	Resistor	330kΩ, ¼W, 5%	1	
R5-R8	Resistor	2.2kΩ, ¼W, 5%	4	
C1	Electrolytic capacitor	1μF, 16V	1	
C2, C3	Ceramic disc capacitor	0.33μF	2	
S1, S2	Switch	Push-to-on	2	
S3, S4	Switch	On/off	2	
J1	Connector	2-pin	1	
CON1-CON17	Connector	2-pin	17	
-	Jumper wires	-	As needed	
APPL1-APPL16	Appliance	230V AC, with 2-pin connector	16	Not included
-	Power supply	5V adaptor	1	Not included
Estimated total				**1100**

Table A

Construction

Assemble the circuit on either a custom-designed PCB or a general-purpose PCB. Connect terminals AP1 through AP16 of the control unit to the corresponding AP1' through AP16' connections on the relay and appliance units. Ensure you also connect the common 5V and GND lines. Once assembly is complete, house the components in the recommended enclosure and power on the system for testing.

Testing

EFY note. Before connecting any 230V AC load, test the circuit with 5V DC only.

1. Check that the 5V supply voltage is correct and that there is no short circuit.
2. Power on the circuit and check that the control section works.
3. Keep S3 and S4 off.
4. Press S1 and S2 and check that the LEDs count up and down correctly.
5. Check that LED2 to LED5 show the selected channel (refer to the table).

For example, to enable APPL16:

OUTPUT STATUS OF DECODER LEDS

Appliance	Decoder Output (A3 A2 A1 A0)	Connect	Transistor/Relay	Appliance on
AP1	0000	AP1 to AP1′	T1 ON, RL1 energised	APPL1
AP2	0001	AP2 to AP2′	T2 ON, RL2 energised	APPL2
AP3	0010	AP3 to AP3′	T3 ON, RL3 energised	APPL3
AP4	0011	AP4 to AP4′	T4 ON, RL4 energised	APPL4
AP5	0100	AP5 to AP5′	T5 ON, RL5 energised	APPL5
AP6	0101	AP6 to AP6′	T6 ON, RL6 energised	APPL6
AP7	0110	AP7 to AP7′	T7 ON, RL7 energised	APPL7
AP8	0111	AP8 to AP8′	T8 ON, RL8 energised	APPL8
AP9	1000	AP9 to AP9′	T9 ON, RL9 energised	APPL9
AP10	1001	AP10 to AP10′	T10 ON, RL10 energised	APPL10
AP11	1010	AP11 to AP11′	T11 ON, RL11 energised	APPL11
AP12	1011	AP12 to AP12′	T12 ON, RL12 energised	APPL12
AP13	1100	AP13 to AP13′	T13 ON, RL13 energised	APPL13
AP14	1101	AP14 to AP14′	T14 ON, RL14 energised	APPL14
AP15	1110	AP15 to AP15′	T15 ON, RL15 energised	APPL15
AP16	1111	AP16 to AP16′	T16 ON, RL16 energised	APPL16

Table B

Note: *Always connect 5V to 5V′ and GND to GND′. Use switches S3 and S4 to turn the appliances on or off.*